UDiTaS run 261030 - library QC

Run 261030_VH00889_534_AAJLG3JM5 - NextSeq 2000, P1, 301 / 8 / 18 / 301 - 8 libraries

Two problems in this run. Most of the library is not genomic insertion junctions, and the eight libraries are far from evenly represented. Both are library-prep issues rather than sequencing issues: the run itself completed normally and base quality is fine.

CLUSTERS SAMPLED	ASSIGNED TO A LIBRARY	TOO SHORT TO MAP	LONG ENOUGH TO MAP
10.31M	82.2%	67.4%	1.5%
3 of 32 tiles	both indexes matched	under 30 nt of sequence	100+ nt of insert

Finding 1 - the library is donor-proximal tagmentation

Tagmentation landed only a few bases past the LE/RE primer, inside the transposon end, so the read reaches the Tn5 adapter before it reaches genomic DNA. 67% of assigned reads carry under 30 bases between the primer and the mosaic end, which is too little to place anywhere in the human genome. Short amplicons also outcompete long ones during PCR, which is why they came to dominate.

The evidence that these are donor and not genome: the same few sequences dominate both the FT and the 293 libraries, at nearly identical frequencies. Two different cell lines cannot share genomic flanking sequence. Those sequences are listed on page 4.

Finding 2 - yield is uneven, and the LE arm came out low

The libraries span 40-fold between the largest and the smallest. Two separate things are happening. The LE arm returned 23% of assigned reads against 77% for the RE arm, so roughly a third as much across the board. Separately, FT_501_253_702 failed almost completely, at 0.56% of assigned reads, which is about ten-fold below the other LE libraries and points at that one reaction rather than at the tagmentation step.

What is still usable

Very little. Only 1.5% of assigned reads carry 100 or more bases of sequence before the Tn5 adapter, which is the minimum worth handing to an aligner. Of those, 9% to 44% are single-base no-signal reads, and most of the rest are donor plasmid: the commonest long sequences are the donor repeat on the RE side and the donor end running into an AscI site and vector backbone on the LE side, and they appear at the same frequency in both cell lines. Reads that are long, varied and not shared between the two lines come to about 0.45% of clusters, an upper bound of roughly half a million reads run-wide, spread over two cell lines and two primer arms. That is too thin for an off-target survey, and what fraction of it is genuine genomic junction cannot be settled without the donor plasmid sequence.

Method: index and read cycles were decoded directly from the run folder's per-cycle base-call files, surface 1, tiles 1101-1103, with no conversion step. Reads were assigned to a library by exact or 1-mismatch match on both indexes. All percentages are of that tile sample, not of the whole run.

Figure 1 - reads per library

Reads whose i5 and i7 both matched the sample sheet, coloured by primer arm. The RE libraries take the four largest shares.

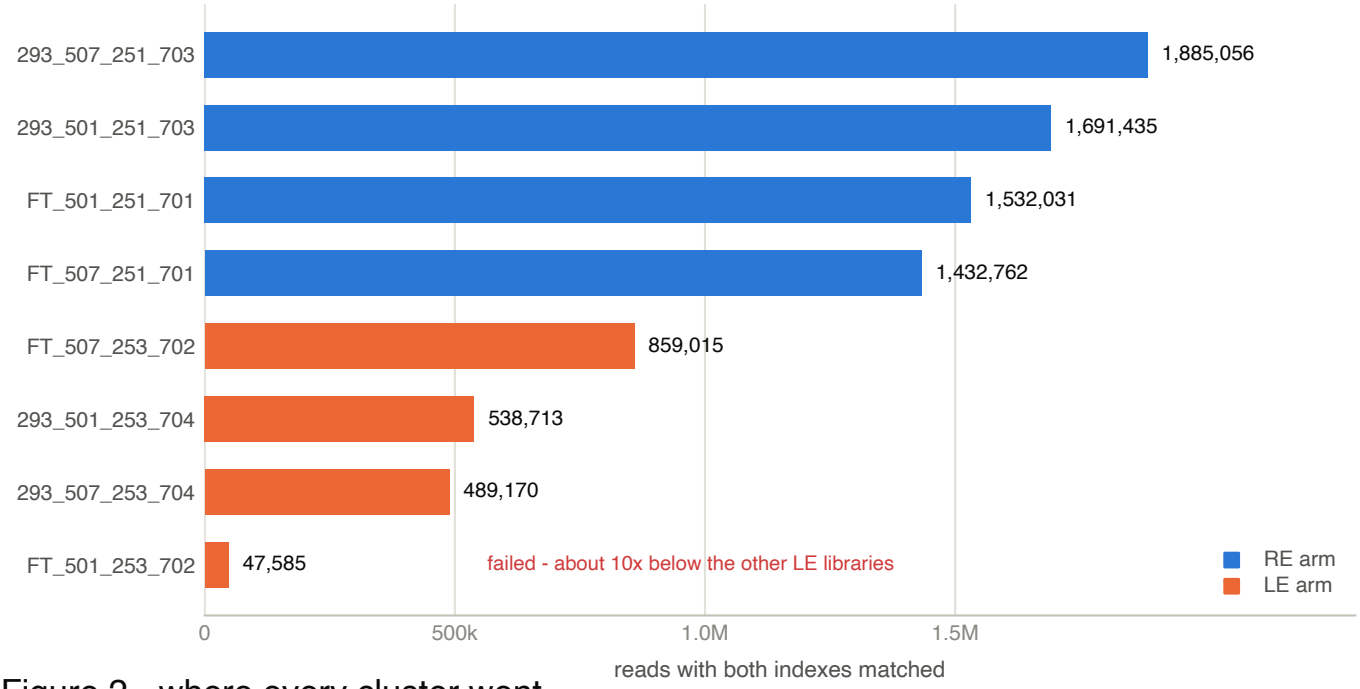
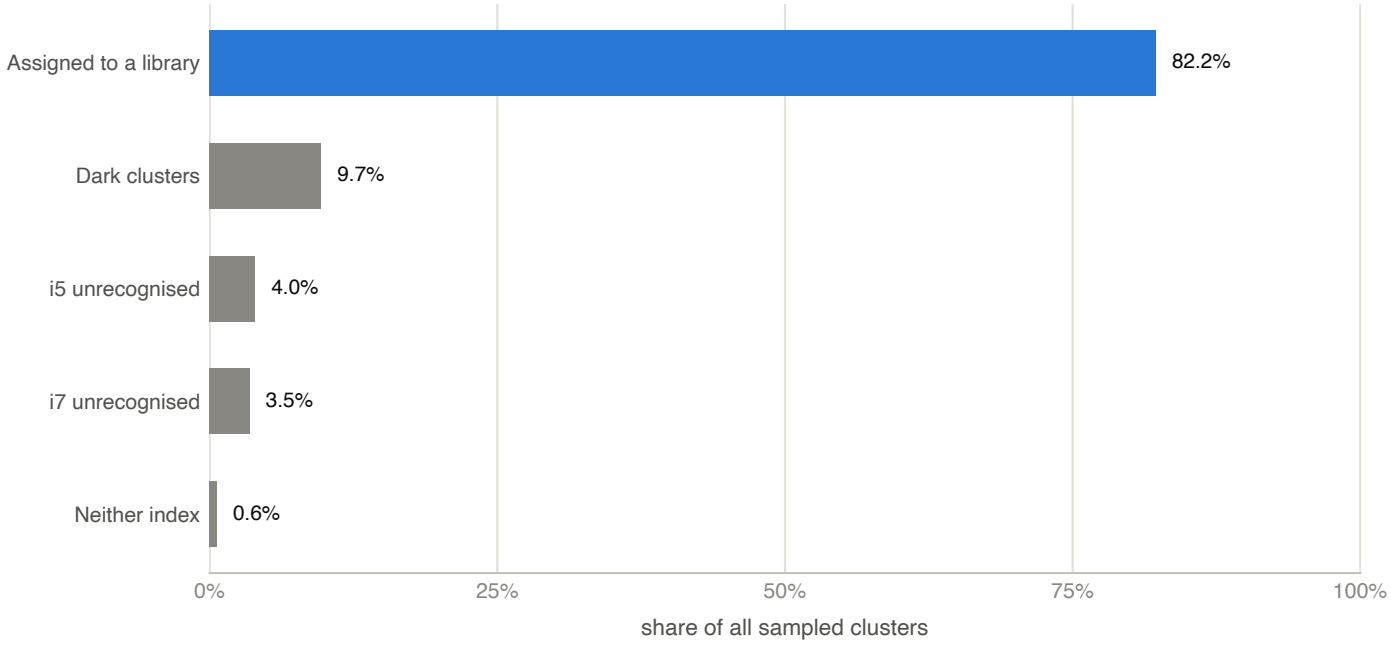


Figure 2 - where every cluster went

Share of all sampled clusters. Dark clusters are wells with no signal in the index read; unrecognised indexes are reads whose barcode matched nothing in the sample sheet within one mismatch.



Of 10,306,025 clusters, 8,475,767 reached a listed library. The RE arm holds 6,541,284 of those and the LE arm 1,934,483.

Figure 3 - fate of the assigned reads

Each library's reads grouped by how much sequence sits between the primer and the Tn5 mosaic end. Only the darkest two bands are long enough to map. The LE libraries are worse than the RE libraries: their products are shorter still. The small remainder to 100% is dark clusters.

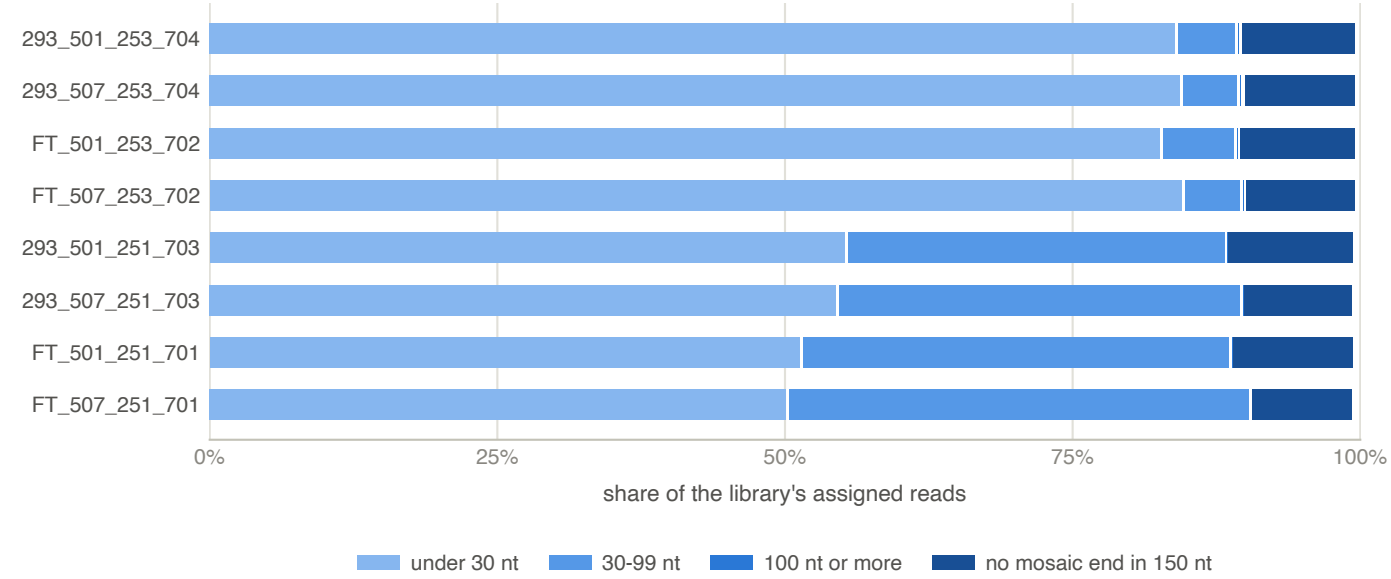


Figure 4 - sequence between primer and mosaic end

Share of reads that contained a mosaic end, by how many bases preceded it. The RE arm peaks under 20 bases with a second group near 50; the LE arm is almost entirely under 30. A usable junction read would sit far to the right of this range. The adapter is detected with a mismatch budget, so adapters carrying sequencing errors are counted here too.

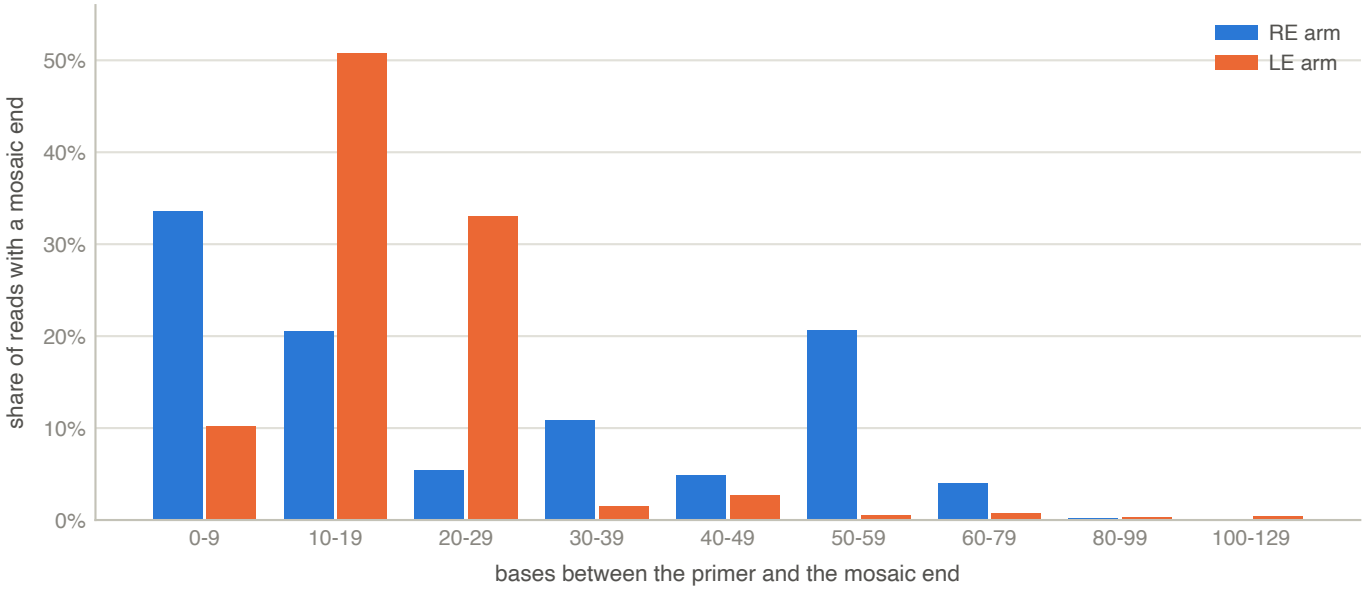


Figure 5 - where the fixed donor sequence ends

Commonest base at each position past the primer, measured only on the long reads. A value near 1 means every read agrees, so the sequence is fixed; the dashed line marks what unrelated genomic sequence would look like. Agreement falls away once the read passes the transposon end, around 30 bases in, and rises again past 100 bases where reads run into the adapter. The fall is why these reads are worth keeping; the rise is where they end.

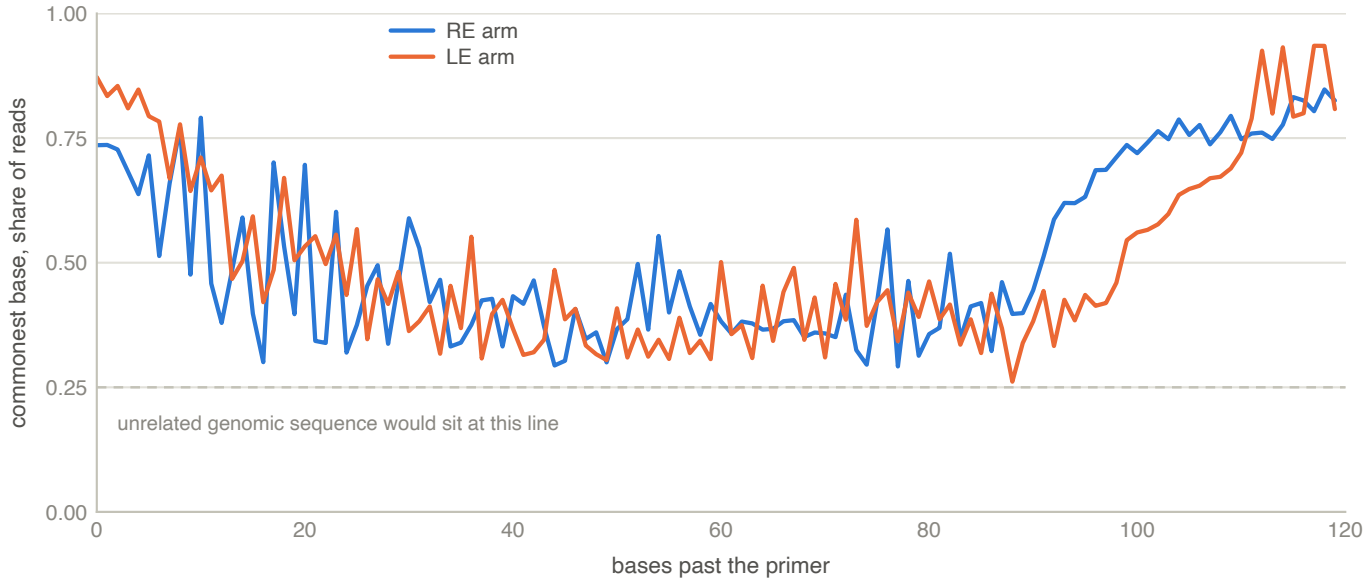


Table 1 - the dominant products, side by side

Most common sequences between primer and mosaic end, per library. Read down the sequence column: FT and 293 share them exactly, at nearly the same frequencies. Genomic flanking sequence could not do this.

Library	Arm	#	Sequence after the primer	nt	Share
293_501_253_704	LE	1	CAAAGTTATGGGT	13	26.6%
	LE	2	CAAAGTTATGGGTAAAGTCACAATC	25	11.2%
	LE	3	CAAAGTTATGGGTAAAGTCACAATCC	26	3.9%
293_507_253_704	LE	1	CAAAGTTATGGGT	13	25.5%
	LE	2	CAAAGTTATGGGTAAAGTCACAATC	25	12.2%
	LE	3	CAAAGTTATGGGTAAAGTCACAATCC	26	4.1%
FT_501_253_702	LE	1	CAAAGTTATGGGT	13	26.1%
	LE	2	CAAAGTTATGGGTAAAGTCACAATC	25	9.7%
	LE	3	CAAAGTTATGGGTAAAGTCACAATCC	26	3.7%
FT_507_253_702	LE	1	CAAAGTTATGGGT	13	24.2%
	LE	2	CAAAGTTATGGGTAAAGTCACAATC	25	13.3%
	LE	3	CAAAGTTATGGGTAAAGTCACAATCC	26	4.5%
293_501_251_703	RE	1	TGTAG	5	15.6%
	RE	2	TGTAGTGTTGTTGACGGCAAAGTTATGGTTCCAGTAACG~	53	9.1%
	RE	3	TGTAGTGTTG	10	7.5%
293_507_251_703	RE	1	TGTAG	5	15.4%
	RE	2	TGTAGTGTTGTTGACGGCAAAGTTATGGTTCCAGTAACG~	53	9.3%
	RE	3	TGTAGTGTTG	10	7.2%
FT_501_251_701	RE	1	TGTAG	5	15.4%
	RE	2	TGTAGTGTTGTTGACGGCAAAGTTATGGTTCCAGTAACG~	53	10.6%
	RE	3	TGTAGTGTTG	10	7.0%
FT_507_251_701	RE	1	TGTAG	5	15.1%
	RE	2	TGTAGTGTTGTTGACGGCAAAGTTATGGTTCCAGTAACG~	53	12.1%
	RE	3	TGTAGTGTTG	10	6.4%

Appendix - the numbers behind the figures

Every figure in this report is drawn from these tables, so each chart can be read as numbers as well.

Per library

Library	Arm	i5	i7	Reads	Share	<30nt	30-99	noME	Donor
293_507_251_703	RE	AAGGAGTA	AAGACGGA	1,885,056	22.24%	54.7%	35.1%	9.7%	42.2%
293_501_251_703	RE	TAGATCGC	AAGACGGA	1,691,435	19.96%	55.5%	33.0%	11.1%	39.9%
FT_501_251_701	RE	TAGATCGC	AGCGGAAT	1,532,031	18.08%	51.6%	37.3%	10.7%	44.8%
FT_507_251_701	RE	AAGGAGTA	AGCGGAAT	1,432,762	16.90%	50.4%	40.2%	9.0%	46.9%
FT_507_253_702	LE	AAGGAGTA	GATCATGC	859,015	10.13%	84.8%	5.0%	9.7%	48.0%
293_501_253_704	LE	TAGATCGC	CGAGTCCT	538,713	6.36%	84.2%	5.2%	10.1%	42.1%
293_507_253_704	LE	AAGGAGTA	CGAGTCCT	489,170	5.77%	84.6%	4.9%	9.9%	44.8%
FT_501_253_702	LE	TAGATCGC	GATCATGC	47,585	0.56%	82.9%	6.4%	10.2%	37.9%

Donor is the share of reads whose first 20 bases past the primer match the donor sequence within 2 mismatches. It undercounts, because a read whose insert is shorter than 20 bases runs into the adapter inside that window.

Cluster assignment

Category	Clusters	Share
Assigned to a library	8,475,767	82.24%
Dark clusters	999,023	9.69%
i5 unrecognised	407,685	3.96%
i7 unrecognised	359,071	3.48%
Neither index	64,479	0.63%

Insert length, share of reads with a mosaic end

Bases before the mosaic end	RE arm	LE arm
0-9	33.6%	10.2%
10-19	20.5%	50.7%
20-29	5.4%	33.0%
30-39	10.8%	1.5%
40-49	4.8%	2.6%
50-59	20.6%	0.4%
60-79	4.0%	0.7%
80-99	0.2%	0.3%
100-129	0.1%	0.3%

Generated from qc_stats.json by make_report.py. Run 261030_VH00889_534_AAJLG3JM5, 3 tiles, 10,306,025 clusters.